SYSTEM DESIGN

Circuit Breakers Visual Guide

Fail fast instead of piling on:
states, thresholds, fallbacks, and bulkheads.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Cascading Failures

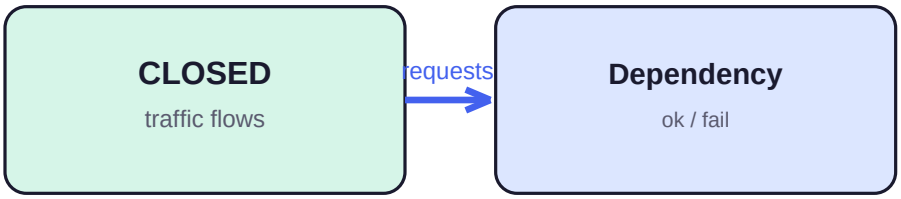
A slow dependency turns every caller into a queue of doomed threads.

The caller's threads block waiting; new requests queue behind them. One slow service kills the healthy ones.

STEP 2/6

CLOSED: Normal Operation

Requests flow; the breaker counts failures in a rolling window.



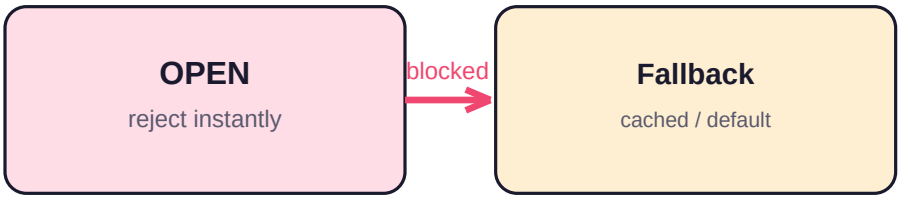
Failure counter: 5 failures in 10s (or 50% of 20 requests) -> TRIP the breaker.

- Thresholds need volume minimums: 2 failures out of 2 requests should NOT trip (noise).
- Rolling window (not cumulative): old failures age out; recovery is possible.

STEP 3/6

OPEN: Fail Fast

The breaker rejects calls immediately -- no doomed threads.

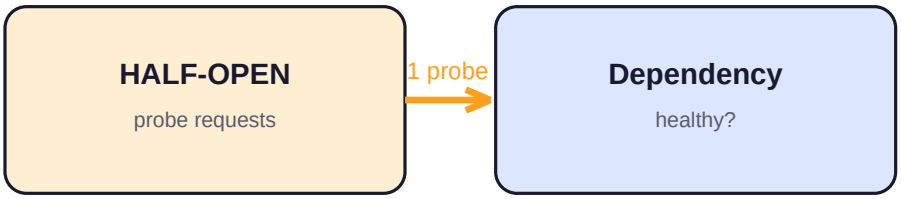


- Fail fast in milliseconds instead of waiting out a 30s timeout -- threads stay free.
- Serve the fallback: stale cache, default value, degraded feature -- 'something' beats 'nothing'.
- Stay open for a cooldown (e.g. 30s): hammering a sick service during recovery kills it again.

STEP 4/6

HALF-OPEN: Probing Recovery

After cooldown, let a trickle through to test the waters.



Probe succeeds -> CLOSED (resume). Probe fails -> OPEN (another cooldown).

Limit probes to ONE at a time: ten half-open callers each sending a probe is still a stampede.

STEP 5/6

Fallbacks: Graceful Degradation

Decide in advance what 'broken but useful' looks like.

Cached

serve last-known-good data with a 'stale' badge

Default

empty recommendations, static content -- the page still renders

Degraded

disable the feature flag; core flow keeps working

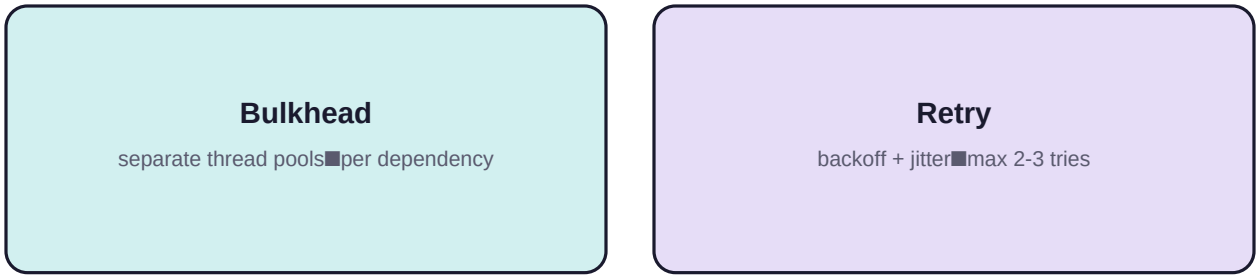
Fail loudly

when there is no safe default: clear error, fast

STEP 6/6

Bulkheads + Retry Discipline

Breakers stop the pile-up; bulkheads and retries stop the spark.



- Bulkhead (the ship metaphor): a flooded compartment cannot sink the ship -- cap each dependency's threads.
- Retry ONLY idempotent ops, with exponential backoff AND jitter (no synchronized retry storms).
- Retry budgets: cap retries at ~10% of traffic so a sick dependency is not drowned in retries.
- Layer them: retry (brief) -> breaker (sustained) -> bulkhead (isolation) -> fallback (UX).

STEP ✓I✓

Key Takeaways

Circuit Breakers — Remember

- Cascading failures happen when callers queue behind a slow dependency -- breakers fail fast instead.
- CLOSED counts failures in a rolling window; trip on rate + minimum volume, not on noise.
- OPEN rejects instantly and serves fallbacks; HALF-OPEN probes with a single request before resuming.
- Design fallbacks up front: cached, default, degraded, or a fast honest error.
- Combine with bulkheads (isolated pools) and jittered, budgeted retries for full resilience.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev